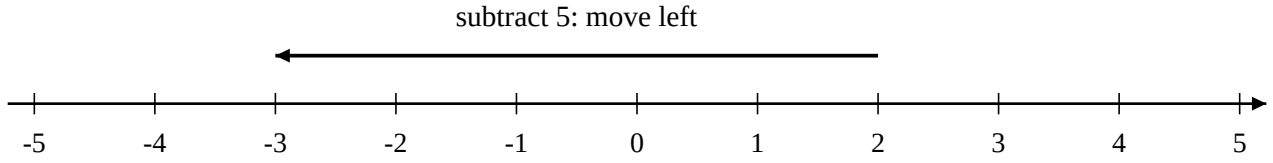
Numbers, Decimals, and Fractions

1. Negative numbers: move on the number line



2 - 5 = -3 Subtracting a negative means add: **4 - (-3) = 4 + 3 = 7**.

Multiply/divide: same signs → positive; different signs → negative.
(-3) × (-2) = 6 (-12) ÷ 3 = -4

Practice: A. 9 + (-14) B. 2 - (-5) C. -18 ÷ 6

2. Decimals: line up places; estimate the answer

Subtract

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  6.00
- 2.75
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  3.25
```

Multiply

1.2 × 0.4
12 × 4 = 48
Two decimal places:
0.48

Divide

7.2 ÷ 0.6
Shift both decimals:
72 ÷ 6 = **12**

Practice: D. 5.4 - 1.85 E. 0.3 × 8 F. 4.5 ÷ 0.5

3. Fractions are division

$\frac{3}{4} = 3 \div 4 = 0.75$



3 of 4 equal parts

Add: use the same denominator. 1/2 + 1/4 = 2/4 + 1/4 = 3/4

Multiply tops and bottoms. (2/3) × (3/4) = 6/12 = 1/2

Divide: flip the second fraction. (3/4) ÷ (1/2) = (3/4) × (2/1) = 1.5

A fraction bar groups the entire top: **(9 - 3)/2 = 6/2 = 3**.

Practice: G. 3/5 as a decimal H. (9 - 3)/2

I. (1/2) × (2/3) J. (2/3) ÷ (1/3)

Check

A -5; B 7; C -3; D 3.55; E 2.4; F 9; G 0.6; H 3; I 1/3; J 2.

Equations, Order, and Units

1. Solve an equation: do the same thing to both sides

x + 7 = 19

↓ subtract 7 from both sides

x = 12

Check: 12 + 7 = 19.

4x = 18

↓ divide both sides by 4

x = 4.5

Check: 4 × 4.5 = 18.

Undo addition with subtraction. Undo multiplication with division.
For x/5 = 3, multiply both sides by 5: **x = 15**.

Practice: K. x + 8 = 15 L. 6x = 21 M. x/4 = 2.5

2. Substitute, then follow the order

Parentheses → exponents → multiply/divide → add/subtract.
Work left to right within each multiplication/division or addition/subtraction step.

$$a = \frac{2 - 8}{3} \longrightarrow \frac{-6}{3} \longrightarrow \mathbf{-2}$$

Do not enter 2 - 8 ÷ 3; that loses the parentheses.
In acceleration, divide the change in velocity by the elapsed time.

Practice: N. (4 - 10)/2 O. 3 + 0.5 × 8

3. Convert units before calculating

minutes —(× 60)→ **seconds**

1.5 min × 60 = 90 s

grams —(÷ 1,000)→ **kilograms**

750 g ÷ 1,000 = 0.75 kg

1 km = 1,000 m. For m/s, use meters and seconds. For F = ma, use kilograms.
m/s² means “meters per second, each second.” It describes acceleration.

Practice: P. 2.5 min in seconds Q. 400 g in kilograms

R. 0.8 km in meters

Finish: round only at the end. Include the unit and direction when needed.

Check

K 7; L 3.5; M 10; N -3; O 7; P 150 s; Q 0.4 kg; R 800 m.